

ANKUR GUPTA

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EDUCATION AND TRAINING

2012-17: Massachusetts Institute of Technology (MIT)
PhD, Chemical Engineering, M.S., Chemical Engineering Practice
Thesis Adviser: Prof. Patrick S. Doyle and Prof. T. Alan Hatton

2008-12: Indian Institute of Technology (IIT) Delhi
B.Tech, Chemical Engineering, President's Gold Medal
Thesis Adviser: Prof. Shantanu Roy

RESEARCH AND PROFESSIONAL EXPERIENCE

2021 - Present: University of Colorado, Boulder
Assistant Professor, Chemical and Biological Engineering Department (ChBE)
(by courtesy) Applied Mathematics Department
(by courtesy) Materials Science and Engineering Program
Principal Investigator: Laboratory of Interfaces, Flow and Electrokinetics (LIFE)

2017-2020: Princeton University
Postdoctoral Research Associate, Mechanical and Aerospace Engineering
Mentor: Prof. Howard A. Stone

RESEARCH IMPACT SNAPSHOT

Scholarly impact	Google Scholar: 4,000+ citations; <i>h</i> -index: 28; 60 peer-reviewed research articles and 3 book chapters; co-editor of one book
Independent output	Since joining CU Boulder, 37 peer-reviewed articles published, 1 manuscript under review, and 2 book chapters under review, including 21 publications where a trainee is the first author and I am the corresponding author Corresponding-author publications in PNAS, Matter, Science Advances, and Journal of Fluid Mechanics, among others
Major recognitions	NSF CAREER; AFOSR YIP; Lyklema Early Career Award; Soft Matter Lectureship; C&EN Talented Twelve; AIChE 35 Under 35
Research support	Secured >\$2.08M extramural funding (total costs; \$1.23M as PI; \$0.85M as co-PI; \$1.33M lab share)
Invited talks	35 invited talks, including Area Plenary (AIChE Fluid Mechanics), invited speaker at APS Global, ELKIN, ACS Fall, and seminars at leading national and international institutions
Visibility	Coverage in The New York Times, National Geographic, The New Yorker, CNN, C&EN, and others; multiple podcasts

TEACHING AND SERVICE SNAPSHOT

Classroom teaching	Taught core undergraduate Fluid Mechanics (CHEN 3200) and graduate Transport Phenomena (CHEN 5210) as sole instructor; aggregate FCQ instructor rating 4.79/5 (departmental mean 4.29); students describe the courses as “the best taught course I’ve taken in a few years” and the instructor as “one of the best STEM teachers I’ve ever had”
Teaching awards	Outstanding Graduate Teaching Award (student-voted 2021–2024; faculty-voted 2022, 2024); Outstanding Undergraduate Teaching Award (student-voted 2024; faculty-voted 2023); College-wide Peebles Innovation in Education Award
Pedagogy scholarship	9 interactive simulations on LearnCheme.com (accessed 6,800+ times); sole-author publication on a class and home problem for electrolyte transport, under review in Chemical Engineering Education; co-author on two ASEE conference papers on the effectiveness of digital experiments and their comparison against physical counterparts
Research supervision	7 PhD students (4 graduated), 1 postdoctoral researcher, and 17 undergraduate researchers; trainees placed at Princeton, NREL, NYU, Exponent Consulting, COMSOL, and top PhD programs
Professional service	Organizing committee, ACS Colloids 2027; symposium and minisymposium organizer and session chair at AIChE, ACS, and APS meetings; elected to the AIChE Fluid Mechanics programming committee, the International Electrokinetic Society Scientific Advisory Board, and the AES Electrophoretic Society Council
Peer review	Reviewed 190+ manuscripts to date (25–30 per year at CU Boulder); grant panels for NSF, and ad hoc reviewer for NSF, DOE, and ACS PRF
Outreach	ACS Webinar panelist (650+ registrants from 59 countries); annual K-12 energy-storage workshops; CU Science Discovery family events

SELECTED AWARDS & HONORS

- ChBE Outstanding Junior Faculty Award (2026)
- Dean’s Faculty Fellowship (2026)
- Soft Matter Lectureship (2025)
- Area Plenary, Fluid Mechanics, AIChE (2025)
- Chemical and Engineering News Talented Twelve (2025)
- Air Force Office of Scientific Research Young Investigator (2025)
- John and Mercedes Peebles Innovation in Education Award (2024)
- Johannes Lyklema Early Career Award in Electrokinetics (2024)
- ChBE Undergraduate Teaching Award (student-voted 2024; faculty-voted 2023)
- ChBE Graduate Teaching Award (student-voted 2021, 2022, 2023, 2024; faculty-voted 2022, 2024)

- American Institute of Chemical Engineers (AIChE) 35 under 35, 2023
- Dream Chemistry Lecture, Physical Chemistry of the Polish Academy of Sciences, 2023
- Session Keynote Speaker, Emulsions, foams and surfactants, ACS Colloids 2023
- NSF CAREER Award, 2023
- Soft Matter Emerging Investigator, RSC Journals, 2023
- Graduates of the Last Decade (GOLD), Alumni Award, IIT Delhi, 2022
- CU Next Award for Innovation in Teaching, 2022
- ACS Petroleum Research Fund, Doctoral New Investigator, 2022
- Defense Advanced Research Project Agency (DARPA) Riser, 2022
- Hugh Hampton Young Fellow, MIT, 2017
- President's Gold Medal, IIT Delhi, 2012

RESEARCH SUMMARY

My group, Laboratory of Interfaces, Flow, and Electrokinetics (LIFE), leverages mathematical tools to tackle interdisciplinary problems across colloidal physics, microhydrodynamics and electrochemistry. Our research aims to illuminate the underlying principles governing these systems which could help advances in material discovery, energy storage devices and miniaturized lab-on-a-chip devices. Collectively, three of the papers led by our group have been profiled in [The New York Times](#), [Science Magazine Newsletter](#), [The New Yorker](#), [CNN](#), [The Knowable Magazine](#), [Yahoo News](#), [CU Boulder Today](#), [The Discover Magazine](#), [Chemical & Engineering News](#), [Popular Mechanics](#), [Interesting Engineering](#), and many others.

PEER-REVIEWED PUBLICATIONS

 [Google Scholar profile](#), 4,000+ citations, h -index=28

Author of 60 peer-reviewed research articles and 3 book chapters (59 articles published, 1 under review; 1 chapter published, 2 chapters under review), and co-editor of one book

Peer-reviewed journal articles from the University of Colorado Boulder

Role key: [graduate trainee](#), [undergraduate trainee](#), [postdoc trainee](#) (names color-coded); ‡ denotes corresponding author, § my PhD advisor, ¶ my postdoctoral advisor; † work initiated during postdoctoral training, published at CU Boulder

Since joining CU Boulder: **37 peer-reviewed articles published, 1 manuscript under review, and 2 book chapters under review, including 21 publications where a trainee (graduate, undergraduate, or postdoctoral) is the first author and I am the corresponding author**

Under review

1. [R. R. Raj](#), J. A. Carter, A. Gupta[‡], C. W. Shields IV[‡], Emergence and exploitation of multiple peaks in microstructure-driven acoustic streaming, *under review*

Published

2. [A. Ganguly](#) and A. Gupta[‡], *Shape-dependence of electrophoretic mobility*, Journal of Fluid Mechanics, accepted, 2026 [[arxiv link](#)], [[blog link](#)] **IF: 3.7**

3. R. Jana, O. Q. Carvalho, R. Bello, P. Chaudhary, J. Hong, A. Hoffman, S. Bare, A. Gupta, V. Alexandrov, K. Stoerzinger[‡], *Disentangling the impact of cation identity on transport and kinetics in nitrate electroreduction to ammonia*, ACS Catalysis, accepted, 2026 **IF: 11.3**
4. G. Origer, R. R. Raj, N. Jarvey, P. N. Romero Zavala, W. A. Smith, and A. Gupta[‡], *Rethinking balance sheets: A Poisson-Nernst-Planck based approach for modeling concentration and flux profiles inside an electrochemical cell*, Chemical Engineering Science, 337, 124600, 2027 [\[link\]](#), [\[blog link\]](#) **IF: 4.1**
5. M. Kelly, E. Dunn, E. Spickermann, J. Gruber, C. Lasalde, P. N. Romero Zavala, E. Lucas, A. Gupta, H. Atwater and W. Smith[‡], *The Missing Piece: Nanometer Scale Imaging to Develop Quantitative Descriptors of Bipolar Membrane Junction Structure*, Scientific Reports, accepted, 2026 **IF: 3.8**
6. P. N. Romero Zavala, A. Ganguly, D. Vigil-Fowler, W. A. Smith and A. Gupta[‡], *Ion transport in charged membranes: Linking electric-field-driven mechanisms to pore size via perturbation analysis*, Langmuir, 42, 19319, 2026 **IF: 3.7**
7. J. Pfeilsticker, E. Coleman, T. Krueger, A. Gupta, and W. Smith[‡], *Spatially mapping the CO₂ alkaline sorbent diffuse microenvironment using operando Raman spectroscopy*, ACS Energy Letters, 11, 3258, 2026 [\[link\]](#) **IF: 18.2**
8. B. Rives, F. Henrique, P. J. Zuk and A. Gupta[‡], *Geometrically induced acceleration for charging dynamics of electrical double layers in a nanopore with sloped walls*, Journal of Physical Chemistry C, 130, 4065, 2026 [\[link\]](#) **IF: 3.2**
9. D. Duong, A. Ganguly, A. Gupta and S. Shin[‡], *Salt-mediated bi-directional propulsion of oil droplets in confined spaces*, Newton (Cell Press), 2, 100405, 2026 [\[link\]](#) **IF: N/A**
10. E. Coleman, and A. Gupta[‡], *A pinch of salt in a pinch of salt: Diffusiophoresis in concentration gradients of three ionic species*, Next Materials, 10, 101431, 2026 [\[link\]](#) **IF: N/A**
11. S. Mirfendereski, and A. Gupta[‡], *Imperfect Turing patterns: Diffusiophoretic assembly of hard spheres via reaction-diffusion instabilities*, Matter (Cell Press), 9, 102513, 2026 [\[link\]](#) **IF: 17.5**
Featured in [The New York Times](#), [Science Magazine Newsletter](#) and 40 other news stories.
12. H. R. Sudhakar, A. Gupta and A. G. Rajan[‡], *Near-Electrode Anion Dehydration and Field-Dependent Dielectric Response Govern Double Layer Capacitance*, Physical Review E, 112, 055503, 2025 [\[link\]](#) **IF: 2.4**
13. A. Shi, S. Mirfendereski, A. Gupta and D.K. Schwartz[‡], *Electrokinetic Nanoparticle Transport in an Interconnected Porous Environment: Decoupling Cavity Escape and Directional Bias*, Proceedings of the National Academy of Sciences, 122, e2514874122, 2025 [\[link\]](#) **IF: 11.1**
14. E. Coleman, and A. Gupta[‡], *Diffusiophoresis in acid-base reaction fronts with and without isoelectric point: When, why and where the particles focus*, Physical Review Fluids, 10, 103701, 2025 [\[link\]](#) **IF: 2.8**
15. R. R. Raj, N. B. Day, N. E. Loomis, E. Cutting, A. Gupta, and C. W. Shields IV[‡], *Transport of adoptive cell transfers with magnetic helical microrobots*, Small, 21, e05946, 2025 [\[link\]](#) **IF: 13.6**
16. B. M. Alessio, and A. Gupta[‡], *Fisher-KPP model with chemotaxis over fractal terrains*, Physical Review E, 112, 024213, 2025 [\[link\]](#) **IF: 2.4**

17. F. Henrique, and A. Gupta[‡], *Parallel Warburg elements describe ionic transport in nanopores*, PRX Energy, 4, 023009, 2025 [\[link\]](#) IF: N/A
18. A. R. Duarte, C. P. Thome, W. S. Hoerdtorfer, C. Praetzel, A. Pellicciotti, A. Gupta, M. A. Bevan, C. W. Shields IV[‡], *Dielectrophoresis-based polarizability measurement (DPM) for predicting induced-charge electroosmotic flows from modified surfaces*, Advanced Functional Materials, 35, 2424557, 2025 [\[link\]](#) IF: 19.0
19. A. Ganguly, S. Roychowdhury and A. Gupta[‡], *Unified mobility expressions for externally-driven and self-phoretic propulsion of particles*, Journal of Fluid Mechanics, 994, A2, 2024 [\[link\]](#) IF: 3.7
20. F. Henrique, P. J. Zuk, and A. Gupta[‡], *A network model to predict ionic transport in porous materials*, Proceedings of the National Academy of Sciences, 121, e2401656121, 2024 [\[link\]](#) IF: 11.1
Featured in [Chemical & Engineering News](#), [Popular Mechanics](#), [Interesting Engineering](#), [MSN](#) and 100 other news stories.
21. A. Shah, S. Pathak, K. Lin, S. Garaj, M. Z. Bazant, A. Gupta and P. S. Doyle^{‡§}, *A universal approximation for conductance blockade in thin nanopore membranes*, Nano Letters, 24, 4776, 2024 [\[link\]](#) IF: 9.1
22. A. Al Harraq, M. Feng, H. Gauri, R. Devireddy, A. Gupta, Q. Sun[‡], and B. Bharti[‡], *Magnetic control of non-magnetic living organisms*, ACS Applied Materials & Interfaces, 16, 17339, 2024 [\[link\]](#) IF: 8.3
23. R. R. Raj, A. Ganguly, C. Becker, C. W. Shields IV and A. Gupta[‡], *Motion of an active bent-rod with an articulating hinge: Exploring mechanical and chemical modes of swimming*, Frontiers in Physics, 11, 2023 (invited article) [\[link\]](#) IF: 1.9
24. A. Ganguly, B. M. Alessio and A. Gupta[‡], *Diffusiophoresis: A novel transport mechanism - Fundamentals, applications, and future opportunities*, Frontiers in Sensors, 4, 2023 (invited article) [\[link\]](#) IF: N/A
25. B. M. Alessio, and A. Gupta[‡], *Diffusiophoresis-enhanced biological Turing patterns*, Science Advances, 9, ead2457, 2023 [\[link\]](#) IF: 11.7
Featured in [CNN](#), [The Knowable Magazine](#), [Yahoo News](#), [CU Boulder Today](#), [The Discover Magazine](#) and 40 other news stories. Selected as a top discovery of CU Boulder 2023.
26. J. G. Lee, C.P. Thome, Z. Cruse, A. Ganguly, A. Gupta, and C. Wyatt Shields IV[‡], *Magnetically locked Janus particle clusters with orientation-dependent motion in AC electric fields*, Nanoscale, 15, 16268, 2023 [\[link\]](#) IF: 5.2
27. A. H. Christensen, A. Gupta, G. Chen, W. S. Peters, M. Knoblauch, H. A. Stone[¶], and K. H. Jensen[‡], *Optimal geometry for surface-enhanced diffusion*, Physical Review E, 108, 045101, 2023 [\[link\]](#) IF: 2.4
28. N. Jarvey, F. Henrique, and A. Gupta[‡], *Asymmetric rectified electric and concentration fields in multicomponent electrolytes with surface reactions*, Soft Matter, 19, 6032, 2023 [\[link\]](#) IF: 2.8
29. A. Seal, U. Tiwari, A. Gupta, and A. G. Rajan[‡], *Incorporating ion-specific van der Waals and soft repulsive interactions in the Poisson-Boltzmann theory of electrical double layers*, Physical Chemistry Chemical Physics, 25, 21708, 2023 [\[link\]](#) IF: 2.9

30. J.G. Lee, [R. R. Raj](#), C. P. Thome, N. B. Day, P. Martinez, N. Bottenus, A. Gupta, and C. Wyatt Shields IV[‡], *Bubble-Based microrobots with rapid circular motions for epithelial pinning and drug delivery*, *Small*, 19, 2300409, 2023 [\[link\]](#) **IF: 13.6**
31. [A. Ganguly](#), and A. Gupta[‡], *Going in circles: Slender body analysis of a self-propelling bent rod*, *Physical Review Fluids*, 8, 014103, 2023 [\[link\]](#) **IF: 2.8**
32. [R. R. Raj](#), C. Wyatt Shields IV, and A. Gupta[‡], *Two-dimensional diffusiophoretic colloidal banding: Optimizing the spatial and temporal design of solute sinks and sources*, *Soft Matter*, 19, 892, 2023 [\[link\]](#) **IF: 2.8**
Selected as a HOT article by editors of Soft Matter, part of a collection for Emerging Investigators in Soft Matter series
33. [F. Henrique](#), P. J. Zuk, A. Gupta[‡], *Impact of asymmetries in valences and diffusivities on the transport of a binary electrolyte in a charged cylindrical pore*, *Electrochimica Acta*, 433, 141220, 2022 [\[link\]](#) **IF: 5.8**
34. [N. Jarvey](#), [F. Henrique](#), A. Gupta[‡], *Ion transport in an electrochemical cell: A theoretical framework to couple dynamics of double layers and redox reactions for multicomponent electrolyte solutions*, *Journal of the Electrochemical Society*, 169, 093506, 2022 [\[link\]](#) **IF: 3.0**
35. [F. Henrique](#), P. J. Zuk, A. Gupta[‡], *Charging dynamics of electrical double layers inside a cylindrical pore: Predicting the effects of arbitrary pore size*, *Soft Matter*, 18, 198, 2022 [\[link\]](#) **IF: 2.8**
36. [B. M. Alessio](#), S. Shim, A. Gupta, H. A. Stone^{‡¶}, *Diffusioosmosis-driven dispersion of colloids: a Taylor dispersion analysis with experimental validation*, *Journal of Fluid Mechanics*, 942, A23, 2022 [\[link\]](#) **IF: 3.7[†]**
37. A. Gupta, A.R. Konicek, M.A. King, [A. Iqtidar](#), M. Yeganeh, H.A. Stone^{‡¶}, *The effect of gravity on the shape of a droplet on a fiber: Nearly axisymmetric profiles with experimental validation*, *Physical Review Fluids*, 6, 063602, 2021 [\[link\]](#) **IF: 2.8[†]**
38. [B.M. Alessio](#), S. Shim, E. Mintah, A. Gupta, H.A. Stone^{‡¶}, *Diffusiophoresis and diffusioosmosis in tandem: Two-dimensional particle motion in the presence of multiple electrolytes*, *Physical Review Fluids*, 6, 054201, 2021 [\[link\]](#) **IF: 2.8[†]**

Peer-reviewed journal articles prior to the University of Colorado Boulder

39. A. Gupta[‡], A. Govind Rajan, Emily A. Carter, H.A. Stone^{‡¶}, *Thermodynamics of electrical double layers with electrostatic correlations*, *The Journal of Physical Chemistry C*, 124, 26830, 2020 [\[link\]](#)
40. A. Gupta[‡], A. Govind Rajan, Emily A. Carter, H.A. Stone^{‡¶}, *Ionic layering and overcharging in electrical double layers in a Poisson-Boltzmann model*, *Physical Review Letters*, 125, 188004, 2020 [\[link\]](#)
41. A. Gupta[‡], P. J. Zuk[‡], H.A. Stone^{‡¶}, *Charging dynamics of overlapping double layers in a cylindrical nanopore*, *Physical Review Letters*, 125, 076001, 2020 [\[link\]](#)
42. A. Gupta, S. Shim, H.A. Stone^{‡¶}, *Diffusiophoresis: From dilute to concentrated electrolytes*, *Soft Matter*, 16, 6975, 2020 [\[link\]](#), *Highlighted in inside front cover*
43. J.L. Wilson, S. Shim, E. Yu, A. Gupta, H.A. Stone^{‡¶}, *Diffusiophoresis in multivalent electrolytes*, *Langmuir*, 36, 7014, 2020 [\[link\]](#)

44. A. Gupta, S. Shim, L. Issah, C. McKenzie, H.A. Stone^{‡¶}, *Diffusion of multiple electrolytes cannot be treated independently: Model predictions with experimental validation*, *Soft Matter*, 15, 9965, 2019 [\[link\]](#)
45. Y. Liu, B. Rallabandi, L. Zhu, A. Gupta, H.A. Stone^{‡¶}, *Pattern formation in oil-in-water emulsions exposed to a salt gradient*, *Physical Review Fluids*, 4, 084307, 2019 [\[link\]](#)
46. A. Gupta, B. Rallabandi, H.A. Stone^{‡¶}, *Diffusiophoretic and diffusioosmotic velocities for mixtures of valence-asymmetric electrolytes*, *Physical Review Fluids*, 4, 043702, 2019 [\[link\]](#)
47. K. Singh, A. Gupta, A. Buchner, F. Ibis, J.W. Pronk, D. Tam, H.B. Eral[‡], *Analysis of centrifugal homogenization and its applications for emulsification & mechanical cell lysis*, *Journal of Colloidal and Interface Science*, 547, 127, 2019 [\[link\]](#)
48. A. Gupta, H. A. Stone^{‡¶}, *Electric double layers: Effect of asymmetry in electrolyte valence on steric effects, dielectric decrement and ion-ion correlations*, *Langmuir*, 34, 11971, 2018 [\[link\]](#)
49. A. Gupta, H. Lee, P.S. Doyle^{‡§}, *Oil recovery from micropatterned triangular troughs during a surfactant flood*, *Langmuir*, 34, 10644, 2018 [\[link\]](#)
50. A.Z.M. Badruddoza, A. Gupta, B.L. Trout, A.S. Myerson, P.S. Doyle^{‡§}, *Low energy nanoemulsions as templates for the formulation of hydrophobic drugs*, *Advanced Therapeutics*, 1, 1700020, 2018 [\[link\]](#)
51. A. Gupta, H. Lee, P.S. Doyle^{‡§}, *Controlled liquid entrapment over patterned sidewalls in confined geometries*, *Physical Review Fluids*, 2, 094007, 2017 [\[link\]](#)
52. A. Gupta, A.Z.M. Badruddoza, T.A. Hatton[§], P.S. Doyle^{‡§}, *A general route for nanoemulsion synthesis using low energy methods at constant temperature*, *Langmuir*, 33, 7118, 2017 [\[link\]](#)
53. H. Lee, A. Gupta, T.A. Hatton[§], P.S. Doyle^{‡§}, *Creating isolated liquid compartments using photopatterned obstacles in microfluidics*, *Physical Review Applied*, 7, 044013, 2017 [\[link\]](#)
54. A. Gupta, V. Narsimhan, T.A. Hatton[§], P.S. Doyle^{‡§}, *Kinetics of change in droplet size during nanoemulsion formation*, *Langmuir*, 32, 11551, 2016 [\[link\]](#)
55. S.G.Lee, H. Lee, A. Gupta, P.S. Doyle^{‡§}, *Site-selective in situ grown carbonate micromodels with tunable geometry, porosity, and wettability*, *Advanced Functional Materials* 26, 4896, 2016 [\[link\]](#)
56. A. Gupta, H.B. Eral, T.A. Hatton[§], P.S. Doyle^{‡§}, *Nanoemulsions: Formation, properties and applications*, *Soft Matter*, 12, 2826, 2016 [\[link\]](#)
57. A. Gupta, H.B. Eral, T.A. Hatton[§], P.S. Doyle^{‡§}, *Controlling and predicting droplet size of nanoemulsions: Scaling relations with experimental validation*, *Soft Matter*, 12, 1452, 2016 [\[link\]](#)
58. G.C.L. Goff, J. Lee, A. Gupta, W.A. Hill, P.S. Doyle^{‡§}, *High-throughput contact flow lithography*, *Advanced Science*, 2, 10, 2015 [\[link\]](#)
59. H. Lee, R.L. Srinivas, A. Gupta, P.S. Doyle^{‡§}, *Sensitive and multiplexed on-chip microRNA profiling in oil-isolated hydrogel chambers*, *Angewandte Chemie International Edition*, 54, 2477, 2015 [\[link\]](#)

60. A. Gupta, S. Roy[‡], *Euler-Euler simulation of bubbly flow in a rectangular bubble column: Experimental validation with radioactive particle tracking*, Chemical Engineering Journal, 225, 818, 2013 [\[link\]](#)

Edited Books

1. G. Bolognesi and A. Gupta (co-editors), *Diffusiophoresis and Diffusioosmosis: Theory, Experiments and Applications*, Royal Society of Chemistry (RSC), chapters submitted April 2026, expected publication December 2026

Book Chapters (3 total)

From the University of Colorado Boulder

1. A. Gupta[‡], *Diffusiophoretic and diffusioosmotic mobilities in electrolyte and non-electrolyte solutions: Detailed derivations across various limits*, in *Diffusiophoresis and Diffusioosmosis: Theory, Experiments and Applications*, edited by G. Bolognesi and A. Gupta, Royal Society of Chemistry, submitted April 2026, expected publication December 2026
2. S. Mirfendereski and A. Gupta[‡], *Diffusiophoresis in biological systems*, in *Diffusiophoresis and Diffusioosmosis: Theory, Experiments and Applications*, edited by G. Bolognesi and A. Gupta, Royal Society of Chemistry, submitted April 2026, expected publication December 2026

Prior to the University of Colorado Boulder

3. A. Gupta[‡], *Nanoemulsions*, invited book chapter in *Nanoparticles for Biomedical Applications: Fundamental Concepts, Biological Interactions and Clinical Applications*, edited by Eun Ji Chung, Lorraine Leon and Carlos Rinaldi, Elsevier, 371, 2020 [\[link\]](#)

FUNDING

Secured over \$2.08M in extramural research funding (total costs; \$1.23M as PI, \$0.85M as co-PI, \$1.33M lab share)

Awarded and Ongoing

1. PI on NSF CAREER Award, *CAREER: Towards a Comprehensive Theoretical Framework to Predict Multiscale and Multicomponent Electrolyte Transport in Porous Media*, \$517,000, Dates: 02/2023 – 01/2028
2. Co-PI (PI: John Falconer, co-PI: Ankur Gupta, Will Medlin) on NSF IUSE, *Collaborative Research: Comparing Effectiveness of Web-based Interactive Digital Experiments to Physical Experiments for Engineering Classrooms*, \$750,000 (my share: \$75,000), Dates: 07/2024 – 06/2027
3. Co-PI (PI: Oana Luca, co-PI: Gupta) on Lab Venture Challenge 2025, *Magnetically Enhanced Electrolysis for Low-Cost Hydrogen Production*, \$100,000 (my share: \$20,000), Dates: 01/2026 – 05/2027
4. PI on AFOSR Young Investigator Program 2025, *Embedding Autonomy in Microscale Systems via Reaction-Diffusion Instabilities*, \$450,000, Dates: 10/2025 – 09/2028

Awarded and completed

1. PI on ACS Petroleum Research Fund Doctoral New Investigator Award, *Nonequilibrium Analysis of Active Droplets Driven by Micellar Solubilization: Effect of In-Situ Change in Droplet Shape and Inter-Droplet Interactions*, \$110,000, Dates: 09/2022 – 08/2024
2. PI on Research & Innovation Seed Grant Program, *Motion of Ellipsoidal Microparticles in Physiological Flows*, \$49,985, Dates: 07/2023 – 12/2024
3. PI on CU Next Award, *Interactive Simulations Based on Neural Networks to Teach Undergraduate Fluid Mechanics*, \$107,000, Dates: 09/2022 – 05/2025

Pending/Planned

1. PI (co-PI: Hind, Bhamla) on Keck Concept Paper, *Predictive Laws of Migration in Complex Gradient Environments Across Scales*, \$1,300,000
2. PI (co-PI: Hind, Bhamla) on ARO Biomathematics, *Predictive Laws of Migration in Complex Gradient Environments Across Scales*, \$720,000, invited for full proposal
3. Co-PI (PI: Goodwin, co-PIs: Cha, Gupta) on NSF, *Effects of Structure and Medium on Cavitation-based Particle Propulsion*, resubmission planned
4. PI on NSF, *Rayleigh-Taylor Instability Induced by Diffusiophoresis Near Ion-Exchange Membranes*, resubmission planned
5. PI (co-PI: Toney) on NSF, *An Integrated Experimental and Theoretical Investigation into Dynamics of Electrochemical Double Layers*, resubmission planned
6. Additional proposals are planned across the three research thrusts; details are described in the Research Statement.

INVITED TALKS

* denotes international presentation

Presentations from University of Colorado Boulder

1. Physics and Astronomy, Clemson University, Clemson, SC, 10/29/2026
2. Chemical and Biomolecular Engineering, University of California Berkeley, Berkeley, CA, 10/21/2026
3. Invited Speaker, AES Electrophoretic Society, SciX, Sparks, NV, 10/07/2026
4. Invited Speaker*, 16th International Symposium on Electrokinetics, The Hague, Netherlands, 06/24/2026
5. Chemical Engineering, California Institute of Technology, Pasadena, CA, 04/16/2026
6. Chemical Engineering, Stanford University, Stanford, CA, 04/04/2026
7. Invited Speaker, APS Global, Denver, CO, 03/16/2026
8. Mechanical Engineering, Yale University, New Haven, CT, 03/04/2026
9. Chemical Engineering, University of Illinois Chicago, Chicago, IL, 01/22/2026
10. Invited Speaker*, AI for Materials and Sustainability, Abu Dhabi, UAE, 12/16/2025
11. Chemical Engineering, Massachusetts Institute of Technology, Cambridge, MA, 11/14/2025
12. Area Plenary, Fluid Mechanics, AIChE 2025, Boston, MA, 11/03/2025

13. Invited Speaker*, International Conference on Micro Nano Fluidics, 11/01/2025 (*remote*)
14. Mathematical Biology, Applied Mathematics, CU Boulder, Boulder, CO, 10/16/2025
15. Chemical Engineering, University of Washington, Seattle, WA, 10/14/2025
16. Emerging Leaders, AES Electrophoretic Society, SciX 2025, Covington, KY, 10/07/2025
17. C&E News Talented 12 symposium, ACS Fall 2025, Washington, DC, 08/18/2025
18. Chemical and Biomedical Engineering, University of Wyoming, Laramie, WY, 03/24/2025
19. Invited Speaker*, Masters Union, Gurugram, India, 12/09/2024
20. Invited Speaker*, Rishihood University, Sonipat, India, 11/25/2024
21. Invited Speaker*, Indian Institute of Technology (IIT) Delhi, New Delhi, India, 11/12/2024
22. Nonlinear Waves Seminar, Applied Mathematics, CU Boulder, Boulder, CO, 10/08/2024
23. Johannes Lyklema Early Career Award Honorary Lecture*, 15th International Symposium on Electrokinetics, Seville, Spain, 09/18/2024
24. Invited Speaker*, 15th International Symposium on Electrokinetics, Seville, Spain, 09/18/2024
25. Invited Speaker, Innovation Day, Science History Institute, Philadelphia, PA, 09/09/2024 (*remote*)
26. Chemical and Biological Engineering, University of Wisconsin, Madison, WI, 04/23/2024
27. Chemical Engineering, Brigham Young University, Provo, UT, 11/30/2023
28. Dream Chemistry Lecture*, Physical Chemistry of the Polish Academy of Sciences, Warsaw, Poland, 07/13/2023
29. ACS Colloids, Keynote in Emulsions, foams and Surfactants, North Carolina State University, Raleigh, NC, 06/06/2023
30. Fluid Mechanics Seminar, Stanford University, Stanford, CA, 05/03/2022
31. National Renewable Energy Laboratory, Golden, CO, 04/08/2022 (*remote*)
32. Physics Colloquium, Los Alamos National Laboratory, Los Alamos, NM, 02/03/2022 (*remote*)
33. Mechanical Engineering, Baylor University, Waco, TX, 10/07/2021 (*remote*)
34. Chemical Engineering, University of Florida, Gainesville, FL, 10/04/2021 (*remote*)
35. Complex Fluids Seminar Series, Carnegie Mellon University, Pittsburgh, PA, 04/16/2021 (*remote*)

Presentations prior to University of Colorado Boulder

36. Soft Matter Coffee Hour (SMATch), Chemical Engineering, Princeton University, Princeton, NJ, 09/16/2020 (*remote*)
37. Chemical Engineering, University of Alberta*, Edmonton, Canada, 04/15/2019
38. Chemical Engineering, National University of Singapore*, Singapore, 04/01/2019
39. Chemical Engineering, Case Western Reserve University, Cleveland, OH, 03/25/2019
40. Chemical Engineering, Michigan State University, East Lansing, MI, 03/05/2019
41. Chemical Engineering, University of Colorado Boulder, Boulder, CO, 02/28/2019
42. Chemical Engineering, Colorado School of Mines, Golden, CO, 02/21/2019
43. Chemical Engineering, University of Oklahoma, Norman, OK, 02/15/2019
44. Chemical Engineering, Indian Institute of Technology (IIT) Delhi*, New Delhi, India, 02/05/2019

45. Chemical Engineering, Indian Institute of Science (IISc) Bangalore*, Bangalore, India, 01/30/2019
46. Chemical Engineering, University of California Davis, Davis, CA, 01/10/2019
47. Chemical Engineering, University of Wisconsin, Madison, WI, 12/05/2018
48. Chemical Engineering, University of Waterloo*, Waterloo, Canada, 08/31/2018
49. Mechanical and Industrial Engineering, Ryerson University*, Toronto, Canada, 08/29/2018
50. Chemical Engineering, Ryerson University*, Toronto, Canada, 08/29/2018
51. Chemical Engineering, McMaster University*, Hamilton, Canada, 08/28/2018
52. Chemical Engineering, McGill University*, Montreal, Canada, 08/24/2018
53. Chemical Engineering, University of Toronto*, Toronto, Canada, 08/08/2018
54. Mechanical Engineering, University of British Columbia*, Vancouver, Canada, 08/02/2018
55. Air Products, Allentown, PA, 03/03/2017
56. Complex Fluids Group, Princeton University, Princeton, NJ, 12/19/2016
57. The Dow Chemical Company, Midland, MI, 10/25/2016
58. Chemical Engineering, Indian Institute of Technology (IIT) Delhi*, New Delhi, India, 03/18/2016

CONTRIBUTED PRESENTATIONS

Presentations from University of Colorado Boulder

† denotes work initiated during postdoctoral training, delivered at CU Boulder

Dates listed as start dates of the conference, list includes talks presented, talks scheduled are not included

1. [B. Rives](#), [F. Henrique](#), P. J. Zuk, and A. Gupta, *Geometric Effects on the Transient Charging of Electric Double Layers in Nanopores*, APS Global Physics Summit 2026, Denver, CO, 03/19/2026
2. [P. N. Romero Zavala](#), [A. Ganguly](#), W. Smith, D. Vigil-Fowler, and A. Gupta, *Pore-Size Impacts on Ionic Conductivity within Ion-Exchange Membranes*, APS Global Physics Summit 2026, Denver, CO, 03/19/2026
3. [S. Mirfendereski](#), [L. Bayer](#), and A. Gupta, *Diffusiophoretic Colloidal Banding and Emergent Hydrodynamic Instability*, APS Global Physics Summit 2026, Denver, CO, 03/19/2026
4. [E. Coleman](#) and A. Gupta, *pH-Dependent Focusing of Diffusiophoretic Particles, With and Without an Isoelectric Point*, APS Global Physics Summit 2026, Denver, CO, 03/20/2026
5. [S. Mirfendereski](#) and A. Gupta, *Diffusiophoretic Assembly of Finite-Size Cells via Reaction–Diffusion Chemical Signal: Imperfect Turing Patterns*, 78th APS-Division of Fluid Dynamics Meeting, Houston, TX, 11/23/2025
6. [S. Mirfendereski](#), [L. Bayer](#), and A. Gupta, *Diffusiophoresis-Driven Exclusion Zone Formation Near an Ion-Exchange Interface and Onset of Hydrodynamic Instability: Experiments and Simulations*, 78th APS-Division of Fluid Dynamics Meeting, Houston, TX, 11/23/2025
7. [E. J. Coleman](#) and A. Gupta, *pH-Dependent Focusing of Diffusiophoretic Particles*, 78th APS-Division of Fluid Dynamics Meeting, Houston, TX, 11/23/2025
8. [S. Mirfendereski](#), A. Gupta, *Diffusiophoretic Exclusion Zone Formation Near Ion-Exchange Membrane and the Onset of Flow Instability*, 2025 Annual AIChE Meeting, Boston, MA, 11/03/2025

9. [S. Mirfendereski](#) and A. Gupta, *Roles of Diffusiophoresis and Electrokinetic Phenomena in Suspensions: From Biological Pattern Formation and Exclusion Zone Dynamics to Suspension Mechanical Responses*, 2025 Annual AIChE Meeting, Boston, MA, 11/03/2025
10. [R. Raj](#), E. Cutting, A. Gupta, and C. Wyatt Shields IV, *Reconciling Gap Distance Predictions via Theoretical and Experimental Analyses of Magnetic Janus Rollers*, 2025 Annual AIChE Meeting, Boston, MA, 11/03/2025
11. [R. Raj](#), N. B. Day, N. E. Loomis, E. Cutting, A. Gupta, and C. Wyatt Shields IV, *Biohybrid Helix–Macrophage Magnetic Microrobots for Cellular Transport through Mucus and Dispersion over Boundaries*, 2025 Annual AIChE Meeting, Boston, MA, 11/03/2025
12. [S. Mirfendereski](#) and A. Gupta, *Imperfect Turing Patterns: Diffusiophoretic Assembly of Hard Spheres via Reaction–Diffusion Instabilities*, 2025 Annual AIChE Meeting, Boston, MA, 11/03/2025
13. [A. Ganguly](#), S. Roychowdhury, [Z. Cruse](#), and A. Gupta, *A Theoretical Study of the Phoretic Motion of Patchy Spherical and Spheroidal Particles through Unified Mobility Expressions*, 2025 Annual AIChE Meeting, Boston, MA, 11/03/2025
14. [R. Raj](#), J. Carter, A. Gupta, and C. Wyatt Shields IV, *Frequency-Dependent Streaming Flows and Motions from Acoustically Powered Microrobots*, 2025 Annual AIChE Meeting, Boston, MA, 11/03/2025
15. [A. Ganguly](#), [R. Raj](#), [P. Suarez Davila](#), [C. Becker](#), and A. Gupta, *Chemical and Mechanical Actuation of Bent-Rod Microparticles Near a Wall*, 2025 Annual AIChE Meeting, Boston, MA, 11/03/2025
16. [E. Coleman](#) and A. Gupta, *pH-Dependent Focusing of Diffusiophoretic Particles*, 2025 Annual AIChE Meeting, Boston, MA, 11/03/2025
17. [P. N. Romero Zavala](#), D. Vigil-Fowler, W. A. Smith, and A. Gupta, *Ion Transport in Charged Nanoporous Networks*, 2025 The Electrochemical Society Meeting, Chicago, IL, 10/12/2025
18. [R. R. Raj](#), A. Gupta, C. W. Shields IV, *Frequency-dependent streaming flows from acoustically actuated bubbles and sharp edges*, 77th APS-Division of Fluid Dynamics, Salt Lake City, UT, 11/24/2024
19. [P. N. Romero Zavala](#), W. A. Smith, A. Gupta, *Reduced-Order Model of Multicomponent Electrolyte Transport in Bipolar Membranes*, 77th APS-Division of Fluid Dynamics, Salt Lake City, UT, 11/24/2024
20. [B. Rives](#), [F. Henrique](#), A. Gupta, *Charging dynamics of electrical double layers in a pore with an axially varying radius: Impact of pore shape and roughness*, 77th APS-Division of Fluid Dynamics, Salt Lake City, UT, 11/24/2024
21. [S. Mirfendereski](#), [E. Coleman](#), A. Gupta, *Particle-Level simulations using diffusiophoresis and cellular automata to create dynamic Turing patterns*, 77th APS-Division of Fluid Dynamics, Salt Lake City, UT, 11/24/2024
22. A. A. Harraq, M. Feng, H. Gauri, A. Gupta, Q. Sun, B. Bharti, *Magnetic Manipulation of Living Organisms without Hybridization*, 2024 AIChE Annual Meeting, San Diego, CA, 10/27/2024
23. [F. Henrique](#), A. Gupta, *Kirchhoff's Laws Get an Upgrade: Double-Layer Dynamics in Pore Networks Described By a De Levie Circuit for an Effective Electrochemical Potential of Charge*, 2024 AIChE Annual Meeting, San Diego, CA, 10/27/2024

24. [S. Mirfendereski](#), A. Gupta, *Merging Turing Patterns and Cellular Automata: Simultaneously Assembling and Evolving Structures Via Diffusiophoresis*, 2024 AIChE Annual Meeting, San Diego, CA, 10/27/2024
25. [F. Henrique](#), P. J. Zuk, A. Gupta, *A Network Model for Ionic Transport in Charged Porous Materials*, ELKIN 2024, Seville, Spain, 09/18/2024
26. [F. Henrique](#), A. Gupta, *Charging dynamics of asymmetric electrolytes in porous media can be represented by magnetically coupled transmission lines*, ACS Fall 2024, Denver, CO, 08/19/2024
27. [A. Ganguly](#), S. Roychowdhury, A. Gupta, *Unified mobility expressions for externally driven and self-phoretic propulsion of particles*, ACS Fall 2024, Denver, CO, 08/19/2024
28. [R. R. Raj](#), A. Gupta, C. W. Shields IV, *Design-driven motion of microrobots powered by acoustic streaming flows*, ACS Fall 2024, Denver, CO, 08/19/2024
29. [S. Mirfendereski](#), [B. M. Alessio](#), [E. Coleman](#), A. Gupta, *Merging Turing patterns and cellular automata: Simultaneously assembling and evolving structures via diffusiophoresis*, ACS Fall 2024, Denver, CO, 08/19/2024
30. [N. Jarvey](#), A. Gupta, *Decomposing total current into capacitive and Faradaic contributions: A theoretical model based on Poisson-Nernst-Planck Equations with Frumkin-Butler-Volmer kinetics*, ACS Fall 2024, Denver, CO, 08/19/2024
31. [P. N. Romero Zavala](#), P. Brimley, W. A. Smith, A. Gupta, *Reduced-order modeling of ion transport in bipolar membranes for electrochemical CO₂ capture and conversion*, ACS Fall 2024, Denver, CO, 08/19/2024
32. [A. Ganguly](#), [R. R. Raj](#), C. W. Shields IV, A. Gupta, *Beyond the scallop theorem: Exploring combined mechanical and chemical propulsion mechanisms of a bent rod actuator*, ACS Fall 2024, Denver, CO, 08/19/2024
33. [F. Henrique](#), P. J. Zuk, A. Gupta, *Kirchhoff's laws get an upgrade: Double-layer dynamics in pore networks described by a de Levie circuit for an effective electrochemical potential of charge*, ACS Fall 2024, Denver, CO, 08/19/2024
34. [R. R. Raj](#), N. B. Day, N. E. Loomis, E. Cutting, A. Gupta, C. W. Shields IV, *Macrophage transport with helical microrobots: Cell attachment, locomotion, and delivery through mucus*, ACS Fall 2024, Denver, CO, 08/19/2024
35. A. Gupta, *Electrolyte transport in electrochemical capacitors: Impact of porous geometry and EDL-redox coupling*, ACS Fall 2024, Denver, CO, 08/19/2024
36. [B. Rives](#), [F. Henrique](#), A. Gupta, *Effects of pore shape and roughness on charging dynamics of electrical double layers*, ACS Fall 2024, Denver, CO, 08/19/2024
37. A. A. Harraq, M. Feng, H. Gauri, R. Devireddy, A. Gupta, Q. Sun, B. Bharti, *Magnetic fields to manipulate non-magnetic living organisms*, 98th American Chemical Society, Colloids and Interface Science, Seattle, WA, 06/23/2024
38. [A. Ganguly](#), S. Roychowdhury, A. Gupta, *A unified mobility expressions for externally driven and self-phoretic propulsion of particles*, 98th American Chemical Society, Colloids and Interface Science, Seattle, WA, 06/23/2024

39. A. Ganguly, R. R. Raj, C. Becker, A. Gupta, *Motion of catalytically active bent rods with an articulating hinge*, 98th American Chemical Society, Colloids and Interface Science, Seattle, WA, 06/23/2024
40. S. Mirfendereski, B. M. Alessio, E. Coleman, A. Gupta, *Diffusiophoresis-Enhanced Turing Patterns: Continuum and Particle-level Simulations*, 98th American Chemical Society, Colloids and Interface Science, Seattle, WA, 06/23/2024
41. N. Jarvey, F. Henrique, A. Gupta, *Asymmetric rectified electric and concentration fields in multicomponent electrolytes with surface reactions*, 98th American Chemical Society, Colloids and Interface Science, Seattle, WA, 06/23/2024
42. F. Henrique, A. Gupta, *Optimization of Pore Shapes for Electrokinetic Flows Produced by Double-Layer Charging*, 98th American Chemical Society, Colloids and Interface Science, Seattle, WA, 06/23/2024
43. F. Henrique, A. Gupta, *Magnetically Coupled Transmission Lines for Double-Layer Charging of Asymmetric Electrolytes in Confinement*, 98th American Chemical Society, Colloids and Interface Science, Seattle, WA, 06/23/2024
44. N. Jarvey, A. Gupta, *A theoretical model to decompose total current into its capacitive and Faradaic contributions for pseudocapacitors*, 98th American Chemical Society, Colloids and Interface Science, Seattle, WA, 06/23/2024
45. B. M. Alessio, A. Gupta, *Diffusiophoresis as a mechanism to study human population migration patterns*, 76th APS- Division of Fluid Dynamics, Washington DC, 11/19/2023
46. B. M. Alessio, A. Gupta, *Diffusiophoresis-enhanced Turing patterns*, 76th APS- Division of Fluid Dynamics, Washington DC, 11/19/2023
47. F. Henrique, P. J. Zuk, A. Gupta, *Kirchhoff's Laws Based on Electrochemical Potential of Charge Dictate Double-Layer Charging in Porous Media*, 76th APS- Division of Fluid Dynamics, Washington DC, 11/19/2023
48. N. Jarvey, F. Henrique, A. Gupta, *Asymmetric rectified electric and concentration fields in multicomponent electrolytes with surface reactions*, 76th APS- Division of Fluid Dynamics, Washington DC, 11/19/2023
49. A. Ganguly, S. Roychowdhury, A. Gupta, *Impact of interaction potential lengthscale and surface heterogeneity on phoretic and autophoretic mobilities: Moving beyond the slip velocity approach*, 76th APS- Division of Fluid Dynamics, Washington DC, 11/19/2023
50. R. R. Raj, J. G. Lee, A. Gupta, C. W. Shields IV, *Effect of geometric design on the motion of microrobots due to acoustic streaming flows*, 76th APS- Division of Fluid Dynamics, Washington DC, 11/19/2023
51. B. M. Alessio, R. R. Raj, and A. Gupta, *Diffusiophoresis-enhanced Turing patterns*, 2023 AIChE Annual Meeting, Orlando, FL, 11/06/2023
52. A. Ganguly, R. R. Raj, C. Becker and A. Gupta, *Self-propelling bent rods: Exploring chemical and mechanical modes of swimming*, 2023 AIChE Annual Meeting, Orlando, FL, 11/06/2023
53. F. Henrique, N. Jarvey, P. J. Zuk and A. Gupta, *Modified Kirchhoff's law for electrical-double-layer charging in porous media*, 2023 AIChE Annual Meeting, Orlando, FL, 11/06/2023

54. [A. Ganguly](#), S. Roychowdhury, and A. Gupta, *Phoretic and Self-Phoretic Motion of Microparticles With Arbitrary Interaction Potentials*, 97th American Chemical Society, Colloids and Interface Science, Raleigh, NC, 06/04/2023
55. [A. Ganguly](#) and A. Gupta, *Slender body analysis of a self-propelling bent rod*, 97th American Chemical Society, Colloids and Interface Science, Raleigh, NC, 06/04/2023
56. [R. R. Raj](#), J. G. Lee, A. Gupta, and C. W. Shields IV, *Impact of geometry on the frequency-dependent response of acoustic microrobots*, 97th American Chemical Society, Colloids and Interface Science, Raleigh, NC, 06/04/2023
57. [F. Henrique](#), P. J. Zuk, and A. Gupta, *Effective Kirchhoff's Laws for Double-Layer Charging in Porous Media*, 97th American Chemical Society, Colloids and Interface Science, Raleigh, NC, 06/04/2023
58. [B. M. Alessio](#) and A. Gupta, *Programmable colloidal assembly: Turing patterns induced via diffusiophoresis*, 97th American Chemical Society, Colloids and Interface Science, Raleigh, NC, 06/04/2023
59. [N. Jarvey](#), [F. Henrique](#) and A. Gupta, *AREFs in multicomponent electrolytes with electrochemical reactions due to imbalance in ionic strength*, 97th American Chemical Society, Colloids and Interface Science, Raleigh, NC, 06/04/2023
60. [N. Jarvey](#), [F. Henrique](#) and A. Gupta, *Coupled ionic transport due to double layers and redox reactions: Impact of multiple ions, background electrolytes, and Frumkin-Butler-Volmer Kinetics*, 97th American Chemical Society, Colloids and Interface Science, Raleigh, NC, 06/04/2023
61. [R. R. Raj](#), C. W. Shields IV and A. Gupta, *Diffusiophoretic colloidal highways: Optimizing the colloidal banding induced by two-dimensional solute gradients*, 97th American Chemical Society, Colloids and Interface Science, Raleigh, NC, 06/04/2023
62. [A. Ganguly](#), A. Gupta, *To turn or not to turn: Slender body analysis for a self-propelling axially asymmetric bent rod*, 75th APS- Division of Fluid Dynamics, Indianapolis, IN, 11/20/2022
63. [R. Raj](#), C. Wyatt Shields IV, A. Gupta, *Rational Design of Two-Dimensional Colloidal Banding*, 75th APS- Division of Fluid Dynamics, Indianapolis, IN, 11/20/2022
64. [N. Jarvey](#), [F. Henrique](#), A. Gupta, *Dynamics of Multicomponent Electrolyte Transport Including the Effects of Electrical Double Layers and Redox Reactions*, 75th APS- Division of Fluid Dynamics, Indianapolis, IN, 11/20/2022
65. [F. Henrique](#), P. J. Zuk, A. Gupta, *Electrical-Double-Layer Charging in a Complex Network of Pores*, 75th APS- Division of Fluid Dynamics, Indianapolis, IN, 11/20/2022
66. A. Christensen, A. Gupta, G. Chen, W. Peters, M. Knoblauch, H. Stone, K. Jensen, *Optimal geometry for surface-enhanced diffusion*, 75th APS- Division of Fluid Dynamics, Indianapolis, IN, 11/20/2022
67. [R. Raj](#), C. Wyatt Shields IV, A. Gupta, *Two-Dimensional Diffusiophoretic Banding of Colloidal Particles*, 2022 Annual AIChE Meeting, Phoenix, AZ, 11/15/2022
68. [A. Ganguly](#), A. Gupta, *Control of Phoretic Self-Propulsion through Particle Geometry: Slender-Body Analysis for an Asymmetric Bent Rod*, 2022 Annual AIChE Meeting, Phoenix, AZ, 11/15/2022

69. [F. Henrique](#), [N. Jarvey](#), A. Gupta, *Transport in Electrochemical Capacitors: Effects of Porous Geometry, Electrolyte Asymmetry, and Redox Reactions*, 2022 Annual AIChE Meeting, Phoenix, AZ, 11/15/2022
70. [A. Ganguly](#), [R. Raj](#), A. Gupta
Impact of Surface Heterogeneity on Diffusiophoresis of Colloids in a Mixture of Electrolytes and Non-electrolytes, 96th American Chemical Society, Colloids and Interface Science, Golden, CO, 07/10/2022
71. [R. Raj](#), C. W. Shields IV, A. Gupta *Programmable Two-dimensional Diffusiophoretic Banding of Colloidal Particles*, 96th American Chemical Society, Colloids and Interface Science, Golden, CO, 07/10/2022
72. [A. Ganguly](#), A. Gupta, *Control of Phoretic Self-Propulsion through Particle Geometry: Slender-body Analysis of an Asymmetric Bent Rod*, 96th American Chemical Society, Colloids and Interface Science, Golden, CO, 07/10/2022
73. [F. Henrique](#), P. J. Zuk, A. Gupta, *Transport of Binary Electrolytes in a Cylindrical Pore: Effects of Overlapping Double Layers and Asymmetry in Ion Valences and Diffusivities*, 96th American Chemical Society, Colloids and Interface Science, Golden, CO, 07/10/2022
74. [N. Jarvey](#), [F. Henrique](#), A. Gupta, *Charging of an Electrochemical Cell: Theoretical Framework to Simulate Coupled Dynamics of Double Layers and Redox Reactions for Arbitrary Number of Ions*, 96th American Chemical Society, Colloids and Interface Science, Golden, CO, 07/10/2022
75. J. G. Lee, [R. R. Raj](#), C. Thome, A. Gupta, C. W. Shields IV, *Bubble-based Acoustic Propellers for Sustained Corticosteroid Delivery in the Bladder*, 96th American Chemical Society, Colloids and Interface Science, Golden, CO, 07/10/2022
76. C. Thome, J. Bendorf, J. G. Lee, A. Gupta, C. W. Shields IV, *Don't Go Breaking My Charge: Induced Charge Electrophoresis of Surface-Modified Janus Particles*, American Chemical Society, Colloids and Interface Science, Golden, CO, 07/10/2022
77. [N. Jarvey](#), [F. Henrique](#), A. Gupta, *Charging of an Electrochemical Cell: Theoretical Framework to Simulate Coupled Dynamics of Double Layers and Redox Reactions for Arbitrary Number of Ions*, 19th U.S. National Congress on Theoretical and Applied Mechanics, Austin, TX, 06/19/2022
78. [F. Henrique](#), P. J. Zuk, A. Gupta, *Charging Dynamics of Electrochemical Capacitors*, 19th U.S. National Congress on Theoretical and Applied Mechanics, Austin, TX, 06/19/2022
79. [N. Jarvey](#), [F. Henrique](#), A. Gupta, *Impact of Faradaic Reactions on the Charging Dynamics of the Electrical Double Layers*, 74th APS - Division of Fluid Dynamics, Phoenix, AZ, 11/21/2021
80. [F. Henrique](#), P. J. Zuk, A. Gupta, *Influence of Relative Debye Length on Electric-Double-Layer Charging Inside a Nanopore*, 74th APS - Division of Fluid Dynamics, Phoenix, AZ, 11/21/2021
81. [F. Henrique](#), A. Gupta, *Charging and Discharging Dynamics of Electrical Double Layers inside Nanopores: From Thin to Overlapping Double Layers*, 2021 Annual AIChE Meeting, Boston, MA, 11/07/2021
82. A. Gupta, A.R. Konicek, M.A. King, [A. Iqtidar](#), M. Yeganeh, H.A. Stone[¶], *The Effect of Gravity on the Shape of a Droplet on a Fiber: Nearly Axisymmetric Profiles with Experimental Validation*, 2021 Annual AIChE Meeting, Boston, MA, 11/07/2021[†]

Presentations prior to University of Colorado Boulder

83. A. Gupta, P. J. Zuk, S. Shim, H. A. Stone, Thick Double Layers: From Energy Storage to Diffusiophoresis, 73rd APS - Division of Fluid Dynamics, Chicago, IL
84. A. Gupta, A. G. Rajan, E. Carter, H. A. Stone, Electrical Double Layers: Predicting Overcharging and Layering of Ions using Continuum Model, 72nd APS - Division of Fluid Dynamics, Seattle, WA
85. A. Gupta, B. Rallabandi, J. L. Wilson, S. Shim, H. A. Stone, Diffusiophoretic Velocity for Mixture of Electrolytes with Asymmetric Ion Valences, 2019 Annual AIChE Meeting, Orlando, FL
86. A. Gupta, H. A. Stone, Electric Double Layers: Effect of Asymmetry in Electrolyte Valence on Finite Ion Size Effects, Dielectric Decrement and Ion-Ion Correlations, 2018 Annual AIChE Meeting, Pittsburgh, PA
87. A. Gupta, A. Z. M. Badruddoza, P. S. Doyle, A General Route for Nanoemulsion Synthesis Using Low Energy Methods at Constant Temperature, 2017 Annual AIChE Meeting, Minneapolis, MN
88. A. Gupta, T. A. Hatton, P. S. Doyle, Nanoemulsion Formation: Controlling and Predicting Droplet Size, 2017 Annual AIChE Meeting, Minneapolis, MN
89. A. Gupta, H. Lee, T. A. Hatton, P. S. Doyle, Controlled Liquid Entrapment through Photo-Patterned Obstacles and Patterned Surfaces, 2017 Annual AIChE Meeting, Minneapolis, MN
90. A. Gupta, T. A. Hatton, P. S. Doyle, Nanoemulsion Formation: Controlling and Predicting Droplet Size, 2016 Annual AIChE Meeting, San Francisco, CA
91. A. Gupta, H. Lee, T. A. Hatton, P. S. Doyle, Controlled Oil Entrapment through Photo-Patterned Obstacles, 2016 Annual AIChE Meeting, San Francisco, CA
92. A. Gupta, T. A. Hatton, P. S. Doyle, Nanoemulsion Formation: Controlling and Predicting Droplet Size, 90th ACS Colloids Meeting, Cambridge, MA
93. A. Gupta, T. A. Hatton, P. S. Doyle, Nanoemulsion Formation: Controlling and Predicting Droplet Size, 90th ACS Colloids Meeting, Cambridge, MA
94. A. Gupta, H. B. Eral, T. A. Hatton, P. S. Doyle, Controlling and Predicting droplet Size of Nanoemulsions, 10th Annual European Rheology Conference, Nantes, France
95. A. Gupta, H. B. Eral, T. A. Hatton, P. S. Doyle, Understanding the Physics of Nanoemulsion Formation, The Society of Rheology 86th Annual Meeting, Philadelphia, PA

RESEARCH SUPERVISION

Graduate (PhD) Student Advisees (7 PhD students advised, 4 graduated)

Sole advisor unless marked co-advised; all advisees are PhD students. Co-advising reflects equal sharing of mentoring and research responsibilities between the listed advisors.

1. Hariharan Ravi Kavitha, PhD (co-advised with Prof. Adam Holewinski; work equally split between groups), University of Colorado Boulder, 2024 - present (*anticipated graduation: Spring 2029*)
Progress: preliminary exam done; first manuscript in preparation

2. Bryce Rives, PhD (sole advisor), University of Colorado Boulder, 2023 - present (*anticipated graduation: Spring 2028*)
Progress: first committee meeting done, second planned for Fall 2026; first-author paper published in J. Phys. Chem. C, another planned soon
3. Peter N. Romero Zavala, PhD (co-advised with Prof. W. A. Smith; work equally split between groups), University of Colorado Boulder, 2023 - present (*anticipated graduation: Spring 2028*)
Progress: first committee meeting done, second planned for Fall 2026; first-author paper published in Langmuir, another planned by the end of the year
4. Ritu Raj, PhD (co-advised with Prof. C. W. Shields IV; work equally split between groups), University of Colorado Boulder, 2021 - 2026
Placement: Associate, Thermal Sciences Practice, Exponent Consulting
5. Arkava Ganguly, PhD (sole advisor), University of Colorado Boulder, 2021 - 2025
Placement: Postdoctoral Research Associate, New York University (PI: Prof. Juan de Pablo)
6. Nathan Jarvey, PhD (sole advisor), University of Colorado Boulder, 2020 - 2024 (defended PhD in August 2024)
Placement: Postdoctoral Research Associate, National Renewable Energy Laboratory
7. Filipe Henrique, PhD (sole advisor), University of Colorado Boulder, 2020 - 2024 (defended PhD in August 2024)
Placement: Postdoctoral Research Associate, Princeton University (PI: Prof. Howard Stone)

Postdoctoral Mentee (1 advised)

1. Siamak Mirfendereski, University of Colorado Boulder, 2024 - 2026 (finished in March 2026)
Placement: Particle Modeling Developer, COMSOL (Boston)

Undergraduate Student Advisees (17 advised at CU Boulder)

1. Chuan Heng, University of Colorado Boulder, 2025-present
2. Sahiti Balaji, University of Colorado Boulder, 2025-present
3. Caitlin Rogers, University of Colorado Boulder, 2025-present
4. Jackson Dunlap, University of Colorado Boulder, 2025-present
5. Ethan Coleman, University of Colorado Boulder, 2024-2025
Placement: Graduate Student, California Institute of Technology
6. Lucas Bayer, University of Colorado Boulder, 2023-2025
Placement: Lead Engineer, Flora Materials
7. Cora Becker, University of Colorado Boulder, 2023-2025
Placement: Quality Engineer, Veolia Water Technologies
8. Grace Origer, University of Colorado Boulder, 2023-2025
Placement: Graduate Student, Oregon State University
9. Paloma Suarez, University of Colorado Boulder, 2024-2025
Placement: Graduate Student, University of California, Santa Barbara
10. Zoe Cruse, University of Colorado Boulder, 2022-2024
Placement: Graduate Student, University of Michigan
11. Sajan Williams, University of Colorado Boulder, 2022-2024
Placement: Chemical Engineer, Synthio Chemicals

12. Ben Alessio, University of Colorado Boulder, 2023
Placement: Graduate Student, Stanford University
13. Eliot Rusley, University of Colorado Boulder, 2022 - 2023
Placement: Civil Engineer, County of Kaua'i
14. Rosby Robinson, University of Colorado Boulder, 2023
Placement: Chemical/Mechanical Engineering Intern, Intel
15. William Steinfert, University of Colorado Boulder, 2022
Placement: Project Engineering Associate, Lockheed Martin
16. Alex Jimenez, University of Colorado Boulder, 2021
Placement: Process Science Associate I, Regeneron
17. Jackson Shropshire, University of Colorado Boulder, 2020
Placement: Associate Project Engineer, Scout Clean Energy
18. Ben Alessio, Princeton University, 2020-2021
19. Azmaine Iqtidar, Princeton University, 2020
20. Cosmin Andrei, Princeton University, 2019
21. Cameron McKenzie, Princeton University, 2018-19
22. Connor Matthews, Princeton University, 2018-19
23. Lisa E. Archibald, MIT, 2016-17
24. Mohammad Alsobay, MIT, 2015
25. Galym Saparbaiuly, MIT, 2015
26. Elezhan Zhakiya, MIT, 2015
27. Robbie Shaw, MIT, 2014-15

AWARDS/HONORS TO ADVISEES

Graduate Advisees

1. May 2026: Peter N. Romero Zavala, ARCS Scholar, ARCS Foundation
2. April 2026: Ritu Raj, Max Peters Outstanding Graduate Student Award, ChBE
3. August 2025: Peter N. Romero Zavala, 2nd place, Poster competition, Front Range Electrochemistry
4. April 2025: Hariharan Ravi Kavitha, Mukhopadhyay Graduate Fellowship, ChBE
5. April 2025: Ritu Raj, Graduate Students' Service to the Department, Volunteering & Professional Development Award, ChBE
6. April 2025: Arkava Ganguly, American Institute of Chemists Graduate Student Award
7. August 2024: Ritu Raj, Presenter at the CU Boulder Innovation in Materials Symposium
8. April 2024: Filipe Henrique, Max Peters Outstanding Graduate Student Award, ChBE
9. April 2024: Ritu Raj, Outstanding Department Teaching Award, ChBE
10. April 2024: Julia Callejon (TA), Outstanding Department Teaching Award, ChBE
11. December 2023: Arkava Ganguly, Teets Family Endowed Doctoral Fellowship, College of Engineering and Applied Science (CEAS)
12. September 2023: Nathan Jarvey, GAANN Fellowship

13. April 2023: Filipe Henrique, Outstanding Department Teaching Award, ChBE
14. June 2023: Nathan Jarvey, Link Energy Fellowship Honorable Mention, Link Foundation
15. June 2023: Nathan Jarvey, GAANN Fellowship
16. April 2023: Ritu Raj, NSF Graduate Research Fellowship
17. January 2023: Ritu Raj, GAANN Fellowship
18. August 2022: Nathan Jarvey, ARCS Scholar, ARCS Foundation
19. July 2022: Filipe Henrique, Langmuir Student Finalist, ACS Colloids
20. April 2022: Arkava Ganguly, Mukhopadhyay Graduate Fellowship, ChBE
21. January 2022: Nathan Jarvey, GAANN Fellowship
22. September 2021: Filipe Henrique, Ryland Graduate Fellowship, CEAS

Undergraduate Advisees

1. May 2026: Caitlin Rogers, Outstanding Senior Award, ChBE
2. April 2026: Ethan Coleman, NSF Graduate Research Fellowship
3. December 2025: Ethan Coleman, Outstanding Undergraduate of the College, CEAS
4. December 2025: Ethan Coleman, Undergraduate Research Award, CEAS
5. December 2025: Ethan Coleman, Academic Engagement Award, CEAS
6. September 2025: Ethan Coleman, Gulf Coast Undergraduate Research Symposium, Rice University
7. September 2025: Ethan Coleman, Future Leaders in Chemical Engineering, NC State
8. April 2025: Cora Becker, Undergraduate Research Award, CEAS
9. November 2024: Paloma Suarez, Poster Award in Material Science, AIChE
10. August 2024: Grace Origer, Student Poster Award in Colloid & Surface Chemistry, ACS Fall
11. August 2024: Paloma Suarez, Second Prize in Materials Category, YSSRP Poster
12. April 2024: Zoe Cruse, NSF Graduate Research Fellowship
13. November 2023: Zoe Cruse, Poster Award in Material Science, AIChE
14. October 2023: Zoe Cruse, Gulf Coast Undergraduate Research Symposium, Rice University
15. April 2023: Ben Alessio, NSF Graduate Research Fellowship
16. April 2023: Zoe Cruse, Undergraduate Research Award, CEAS

Postdoctoral Mentee

1. April 2026: Siamak Mirfendereski, American Institute of Chemists Postdoctoral Award
2. September 2025: Siamak Mirfendereski, Outstanding Postdoc of the Year (Honorable Mention), CU Boulder

TEACHING EXPERIENCE

Engineering Education & Pedagogy Scholarship

Peer-reviewed scholarship on the teaching of transport and electrokinetics

1. A. Gupta, *A class and home problem on electrolyte transport: constant electric field implies electroneutrality, but electroneutrality does not imply a constant electric field*, under review at Chemical Engineering Education, [\[arxiv link\]](#)

2. R. J. Fosbre, F. Rahimi, D. B. Thiessen, J. L. Falconer, J. W. Medlin, A. Gupta, and B. J. Van Wie, *NSF IUSE: Progress in evaluating hands-on learning module implementation with a social cognitive lens and alternative web-based experiments*, ASEE Annual Conference, 2026
3. J. L. Falconer, D. M. Smith, V. S. S. N. Kavatooru, D. B. Thiessen, B. J. Van Wie, R. J. Fosbre, F. Rahimi, A. Gupta, and J. W. Medlin, *Web-based digital experiment in chemical engineering*, ASEE Annual Conference, 2026

Courses Taught

1. Instructor, Transport Phenomena (CHEN5210), 4 credits

University of Colorado Boulder, Fall 2025

Course level: graduate, enrollment: 16

Evaluated as "a phenomenal lecturer," with students noting that the course was "beautifully structured and inculcates interest," and that they walked out of lectures with "a new sense of wonder about how this world works."

2. Instructor, Transport Phenomena (CHEN5210), 4 credits

University of Colorado Boulder, Spring 2025

Course level: graduate, enrollment: 27

Evaluated as "the best graduate professor I have had," praised for being "incredibly organized," "very good at explaining math-heavy questions in a conceptual way," and for creating one of the "most organized and instructive" courses students have taken.

3. Instructor, Fluid Mechanics (CHEN3200), 3 credits

University of Colorado Boulder, Spring 2024

Course level: undergraduate, enrollment: 77

Evaluated as "Probably one of my favorite courses I've taken at CU," "Genuinely one of the best STEM teachers I've ever had," "The most effective aspect of the course was how well Professor Gupta is able to answer questions during lecture. It is not only able to break up the pace of the lecture, but also able to hear other people's thinking/logic on a topic," "Professor Gupta is probably one of the best professors I've had at CU Boulder. I think the biggest reason for this is his ability to create such a good learning environment for his students."

4. Instructor, Transport Phenomena (CHEN5210), 4 credits

University of Colorado Boulder, Spring 2024

Course level: graduate, enrollment: 28

Evaluated as "This class was my favorite course. I think that the materials and information were organized very well," "I feel like out of all of the classes I've ever taken, this is the one that encouraged me the most to actually understand the material rather than just repetitively applying information I wasn't truly in tune with. The material was very interesting, and you presented it in a logical and engaging way," "This was an excellent course. It was highly engaging content that was well-organized and well-taught."

5. Instructor, Fluid Mechanics (CHEN3200), 3 credits

University of Colorado Boulder, Spring 2023

Course level: undergraduate, enrollment: 67

Evaluated as "teaching style to be phenomenal," "lectures are very informational, concise, and delivered really well on behalf of Prof Gupta," "Professor truly knows what he's doing and how to teach, utilizing the perfect mix of lecturing and example problems."

6. **Instructor, Transport Phenomena (CHEN5210), 3 credits**
 University of Colorado Boulder, Fall 2022
 Course level: graduate, enrollment: 29
Evaluated as "One of the best teachers I have had," "This course was taught at what I believe is an impeccable level," "Ankur's teaching style is just fantastic," "I wish every CHEN graduate course could be taught by Ankur," "Honestly the best taught course I've taken in a few years. I can't think of a single thing in the course that I wish was better."
7. **Instructor, Transport Phenomena (CHEN5210), 3 credits**
 University of Colorado Boulder, Fall 2021
 Course level: graduate, enrollment: 18
Evaluated as "Professor Gupta's teaching style was very engaging," "Dr Gupta gives highly effective lectures which engage students," "Ankur is the best professor I have ever had."
8. **Instructor, Transport Phenomena (CHEN5210), 3 credits**
 University of Colorado Boulder, Spring 2021
 Course level: graduate, enrollment: 34
Evaluated as "Ankur Gupta is a phenomenal instructor," "Ankur was the best professor I have ever had," "Professor Gupta is the best example I have seen of teaching effectively in both virtual and hybrid modes."
9. **Guest Lecturer, Advanced Heat and Mass Transfer (CBE505)**
 Princeton University, Spring 2020
 Course level: graduate, enrollment: 25
 Responsibility: developed and delivered 3 lectures on electrokinetics (*delivered remotely due to COVID-19*)
10. **Instructor, Electrokinetics for Energy and the Environment (MAE 559)**
 Princeton University, Fall 2018
 Course level: graduate, enrollment: 20 (including audit, listeners)
 Responsibility: developed and delivered 75% of lectures
11. **Graduate Instructor, Fluid Mechanics (10.301)**
 MIT, Spring 2017 Course level: undergraduate, enrollment: 58
 Responsibility: 40% lectures, 50% recitations
12. **Teaching Assistant, Transport Processes (10.302)**
 MIT, Fall 2014
 Course level: undergraduate, enrollment: 71
13. **Teaching Assistant, Junior Design Course (CHL471)**
 IIT Delhi, Spring 2012
 Course level: undergraduate, enrollment: 120
14. **Instructor of Mathematics and Physics, Vidyamandir Classes**
 Delhi, 2009-11
 Course level: high school, enrollment: 400 (40 × 10)

PROFESSIONAL SERVICE

1. Referee for journals (190+ journal articles reviewed excluding revisions, about 25-30 articles/year)

Journal list includes: Nature Communications, Nature Physics, Angewandte Chemie, Physical Review Letters, Journal of Fluid Mechanics, Advanced Functional Materials, ACS Applied Materials & Interfaces, Langmuir, Soft Matter, Physical Review Fluids, Physical Review E, Physical Review Applied, Journal of Colloid & Interface Science etc.

2. Grant Proposal Reviewer

Research Foundation – Flanders (FWO), Belgium (x2 *ad-hoc*)

ACS Petroleum Research Fund (x4 *ad-hoc*)

National Frontiers in Research Fund, Canada (x2 *ad-hoc*)

National Science Foundation (x5 panels, x2 *ad-hoc*)

Israel Science Foundation (x2 *ad-hoc*)

Department of Energy (x1 *ad-hoc*)

3. Organizer or chair of sessions at scientific meetings

co-Organizer, ACS Colloids, 2027

Judge, Poster session, AI for Materials and Sustainability, 2025

Chair, Young Researchers and Scientists Sessions, AI for Materials and Sustainability, 2025

Chair, Nonlinear and Turbulent Flows and Transport Processes, AIChE 2025

Chair, Interfacial and Non-Newtonian Flows, AIChE 2024

Chair, Interfacial and Non-Newtonian Flows, 2024

Organizer, Interfacial Phenomena & Dynamics in Electrochemical Systems, 4-session minisymposium, ACS Fall, 2024

Chair, Electrokinetics and Microfluidics, ACS Colloids, 2024

Chair, Electrokinetic Transport III, APS DFD, 2023

Chair, Micro/Nano scale Flows: Electrokinetics, APS DFD, 2023

Chair, Interfacial and Nonlinear Flows: Multiphase and Fields, AIChE Annual Meeting, 2023

Judge, Student Poster Session, APS DFD, 2022

Chair, Microfluidic and Microscale Flows, AIChE Annual Meeting, 2022

Organizer, Electrokinetics for Nano- and Microfluidics, 2-day minisymposium, USNCTAM, 2022

Chair, General Aspects for Colloids and Interface, ACS Colloids, 2022

Chair, Interfacial and Nonlinear Flows: Multiphase and Fields, AIChE Annual Meeting, 2021

4. Leadership positions at societies

Councilor, AES Electrophoretic Society, 2026 - present

Scientific Advisory Board, International Electrokinetics Society, 2025 - present

Fluids Programming Committee, AIChE, 2024-2034

SERVICE AT UNIVERSITY OF COLORADO BOULDER

1. Departmental committees and service

Seminar Organization 2025-2026

Regrade Committee 2025-2026

Graduate Committee 2021-present
Teaching Quality Framework Committee, 2021-2023
New Chair Search Committee, 2023
Graduate Student Award Committee, 2021-2023, 2025
Graduate and Postdoc Award Committee Chair, 2026
Outstanding Doctoral Dissertation Committee, 2023
Preliminary Exam Committee, 2021-present

2. Thesis committees

Served on 30 Ph.D. thesis committees at the University of Colorado Boulder

Gesse Roure, Chemical Engineering, 2021 - 2023
Yifeng Mao, Applied Mathematics, 2022 - 2024
Paige Brimley, Chemical Engineering, 2021 - 2024
Laura Herrera, Chemical Engineering, 2021 - 2024
Cooper Thome, Biological Engineering, 2021 - 2024
Nate Schwindt, Chemical Engineering, 2022 - 2025
Katarina Odak, Chemical Engineering, 2022 - 2025
Talaia Alina, Chemical Engineering, 2022 - 2025
Luis Kitsu, Chemical Engineering, 2022 - 2025
Benjamin Rich, Chemistry, 2023
Owen Asaro Lee, Chemical Engineering, 2023 - 2024
Julie Nguyen, Chemical Engineering, 2023 - 2025
Hussain Almajed, Chemical Engineering, 2023 - 2025
Trisha Nickerson, Chemical Engineering, 2023 - 2026
Brandon Oliphant, Chemical Engineering, 2023 - 2026
Kendra Kreienbrink, Materials Science and Engineering, 2024 - 2026
Ashutosh Mishra, Chemical Engineering, 2025 - 2026
Noah Smith, Chemical Engineering, 2023 - present
Ian Wylie, Chemical Engineering, 2024 - present
Collin Kemper, Chemical Engineering, 2024 - present
Rajarshi Chattopadhyay, Chemical Engineering, 2024 - present
Souradeep Roychowdhury, Chemical Engineering, 2024 - present
Timotej Bernat, Chemical Engineering, 2024 - present
Rafael Ferreira de Menezes, Chemical Engineering, 2025 - present
Madelyn Bennett, Chemical Engineering, 2025 - present
Rebecca Beswick, Chemical Engineering, 2025 - present
Varun Awasthi, Chemical Engineering, 2025 - present
Linnea Helenius, Chemical Engineering, 2025 - present
Eva Peurrung, Chemical Engineering, 2025 - present
Jesus Melendez, Chemical Engineering, 2025 - present

OUTREACH ACTIVITIES

1. **Who will win the 2025 Nobel Prize in Chemistry**, Panelist, ACS Webinar ([link](#)), 650+ registrations from 59 countries

2. **Workshop on Energy Storage**, Office of Precollege Outreach and Engagement, Summer 2024, 2025
3. **Podcast speaker**, How animals get their spots, and why they are beautifully imperfect, Quantum Photonic
4. **Podcast speaker**, Knocking On All Doors!, Dil se IIT Delhi Podcast ([link](#))
5. **Podcast speaker**, The Science Behind Your Pets' Beautiful Patterns & other Inspiring Stories ([link](#))
6. **Digital simulations on LearnCheme.com**
 - Bernoulli's pipe flow network, ([link](#))
 - Buoyancy of a floating cube, ([link](#))
 - Contact angle measurement, ([link](#))
 - Flow between three tanks, ([link](#))
 - How fast does a tank empty? ([link](#))
 - Open parachute, ([link](#))
 - Rising 3D-printed quarter, ([link](#))
 - Viscous flow around a rotating rod, ([link](#))
 - Visual velocity profile, ([link](#))